

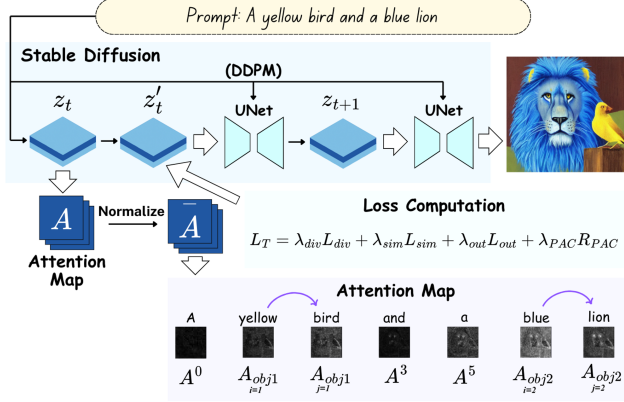


Figure 2. **An overview of our workflow for optimizing the stable diffusion model.** It includes aggregation of attention maps, computation of object-centric attention loss, and updates to z_t .

models, paving the way for more reliable and interpretable generative models.

1. Introduction

Recent text-to-image (T2I) diffusion models have achieved remarkable success in generating high-fidelity, diverse, and realistic images conditioned on textual descriptions [3, 8–10, 13, 16, 20, 27, 31, 33, 35, 36, 38]. Techniques such as classifier-free guidance [15] and improved training strategies [28] have further enhanced their performance. However, handling complex prompts with multiple objects and attributes remains challenging [6, 11, 24, 32, 44]. Diffusion models often misassociate attributes with incorrect nouns, leading to incorrect attribute bindings or omission of certain objects [6, 32, 44].

Challenges in attribute-object alignment are largely attributed to limitations within the models’ attention mechanisms, which operate as black boxes and lack transparency [10, 11, 14, 32, 40]. Without explicit control over how attention is allocated, models struggle with tasks requiring precise modifier-object associations, resulting in semantic inaccuracies and compositional errors in the generated images [2, 4, 14]. Although various methods have attempted to guide attention, such as *Prompt-to-Prompt* editing [14], *Composable Diffusion* models [24], and *Training-Free Structured Diffusion Guidance* [11], issues like attribute misbinding and insufficient generalization persist [6, 44].

In response to these limitations, we propose a novel Bayesian framework to enhance attention mechanisms within text-to-image diffusion models. Our approach leverages the PAC-Bayes framework [25] to allow users to design custom priors over attention distributions, enforcing desired properties such as divergence between different

objects, similarity between modifiers and their respective nouns, minimized attention to irrelevant tokens, and a regularization component for robust generation. By minimizing the Kullback-Leibler divergence between learned attention distributions and custom-designed priors, our framework guides models towards improved attribute-object alignment and balanced attention allocation, effectively addressing limitations in previous approaches.

For instance, Moreover, issues such as object hallucination [19], semantic confusion [12], and context misalignment [23] further exacerbate these challenges.

Our contributions can be summarized as follows:

- We introduce a Bayesian framework that allows for custom priors over attention distributions, enhancing control and addressing the black-box nature of attention mechanisms.
- We formalize the problem within the PAC-Bayes framework [25], providing theoretical generalization guarantees.
- Our method achieves state-of-the-art results on standard benchmarks [6, 11, 32, 44], validating our approach’s effectiveness in improving attribute binding and attention allocation.

2. Related Works

2.1. Stable Diffusion

Stable Diffusion [35] represents a major advancement in text-to-image synthesis, utilizing a latent diffusion model to generate high-resolution images efficiently. By operating in a lower-dimensional latent space [41] learned through an autoencoder [10, 34], Stable Diffusion reduces computational demands compared to pixel-space diffusion models [8, 16, 37]. Cross-attention mechanisms [30, 40] enable Stable Diffusion to align textual and visual information, producing detailed images from textual prompts [22, 27, 36]. Techniques like classifier-free guidance [15] and improved training objectives [28] have further enhanced generation quality and control [43, 47].

However, while these models excel at simple text-to-image tasks, they struggle with prompts involving multiple objects and attributes [6, 11, 32]. Specifically, Stable Diffusion and similar models frequently misbind attributes to incorrect objects or overlook certain objects entirely, which limits their utility for tasks requiring precise modifier-object alignment [5, 42, 44].

2.2. Attention-Based Methods for Attribute-Object Alignment

Several recent methods have been developed to improve attribute-object alignment by manipulating attention mechanisms. Hertz et al. [14] introduced *Prompt-to-Prompt* editing, which enables users to control generation by adjusting